

# Transcribe Audio Files with AI



Natasha Ong

| Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Audio identification | Subtitles | Toxicity detection - new |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------|--------------------------|
| <p><b>Speaker 0:</b> Indicates that Docker couldn't pull a Docker image from Amazon ECR, OK, the specific error is a 4 or 3 forbidden response which usually points to an issue with permissions or access rights. OK, that's what we thought, which is ***. Uh-huh. Since repositories with ECR are private by default, both you and Player A have set up private repositories that no-one else can access until you give someone else specific permissions.</p> <p><b>Speaker 1:</b> Yeah, so what we've set up are private repositories, my friends. Do you see even in the left-hand navigation panel for Maximus, it's images that highlight a thing is under the heading private registry. ***, so what's really happening here is it's so private that no one else, *** no one else can have access until we give it to them specifically. And that's the reason why we can just openly paste the links to our registries in the chat because actually we know that even if you have access to that link, you can't actually get access to the registry itself until we give you specific access. ***.</p> |                      |           |                          |



# Introducing Today's Project!

In this project, we will demonstrate how to use AI to create a transcription for audio and videos. We're doing this project to learn how we can create transcriptions that enable voice commands or improve accessibility for an app!

## Tools and concepts

Services we used were Amazon S3 for video storage and Amazon Transcribe for transcription. Key concepts we learnt include live transcribing, custom vocabularies and filters, Transcribe <> S3's connection.

## Project reflection

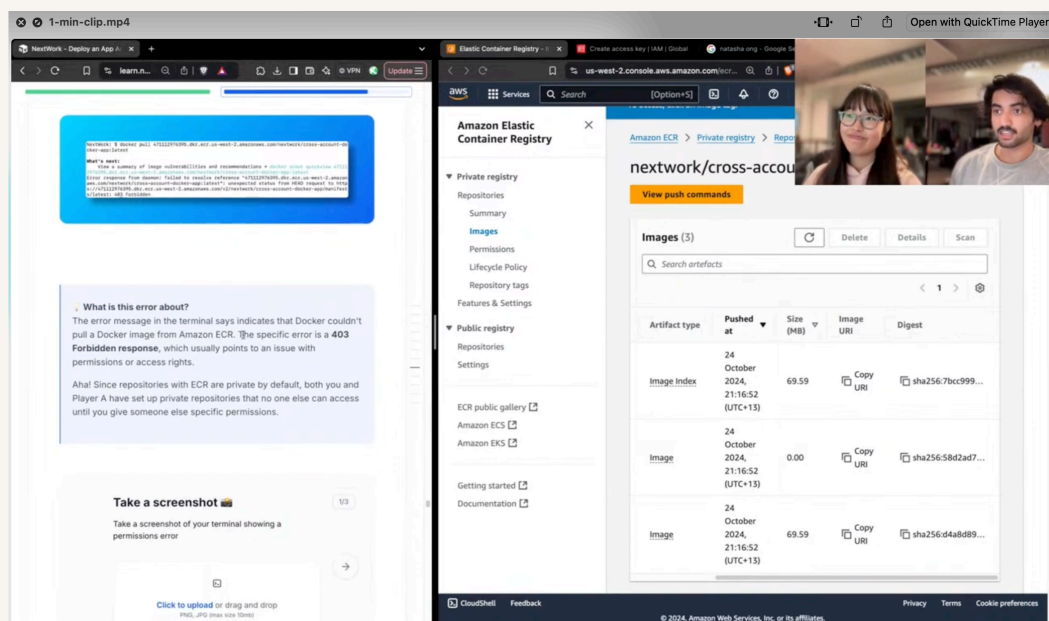
This project took us approximately 1.5 hours including demo time. The most challenging part was creating the custom vocabulary. It was most rewarding to do a live transcript and see the custom vocab + filter at work straight away!

We did this project today to learn about transcription and AI tools like Transcribe! This project helped meet our goals by letting us practise transcription + ways to make it even more accurate.



# S3 and Transcribe

To set up for this project, we're using an S3 bucket to store the video file that we're transcribing. The file we're transcribing is a 1 minute clip of an existing NextWork project demo. This is a great opportunity to provide subtitles to our videos!





# Run A Transcription Job

The steps to run a transcription job include selecting a language and model type; selecting input and output data (i.e. where the video/audio is from, and where the transcription should be stored). Overall, this process took us about 5 minutes.

Amazon Transcribe uses model types to learn how to translate speech to text. You can think of a model type as an instruction or guide that Transcribe uses. Use cases of model types include creating custom models for medical/legal conversations.

We can customise a transcription further with subtitling, which adds subtitles to a video (great for accessibility or translations); and speaker partitioning, which helps to identify multiple speakers in a audio file.

**Vocabulary filtering** [Info](#)  
Vocabulary filtering can remove, mask or tag specified words in the final transcript.

**Filter selection**  
The vocabulary filters shown here are based on your language settings. You can choose up to one vocabulary filter per language. You can also [create a new vocabulary filter](#).

filter-words-filter ▼

**Vocabulary filtering method**  
Use a vocabulary filter to filter vocabularies from your transcript. For example, in the sentence 'The quick brown fox jumps over the lazy dog', you remove the word 'lazy' using the following options.

☒ **Mask vocabulary**  
Example: the quick brown fox jumps over the \*\*\* dog.

☐ **Remove vocabulary**  
Example: the quick brown fox jumps over the dog.

☐ **Tag vocabulary**  
Example: The quick brown fox jumps over the lazy.dog.

**Customisation**

**Customised vocabulary** [Info](#)  
A customised vocabulary improves the accuracy of recognising words and phrases specific to your use case.

**Vocabulary selection**  
The vocabularies shown here are based on your language settings. You can choose up to one vocabulary per language. You can also [create a new vocabulary](#).

nextwork-project-vocab ▼





# Baseline Transcript Review

To start using Amazon Transcribe, we first ran a baseline transcription job, which means a transcription that doesn't use any of the additional features or tools provided by Amazon Transcribe. This lets us compare it with the results of any changes.

While reivewing the baseline transcription, we noticed a few inaccuracies, including typos (e.g. `repositories` spelt as `repositoriesies`), jargon (e.g. `403 Forbidden` transcribed as `4 or 3 forbidden`), filler words (e.g. `um`) and lack of context.

Text

Audio identification

Subtitles

Toxicity detection - new

Indicates that Docker couldn't pull a Docker image from Amazon ECR, OK, the specific error is a 4 or 3 forbidden response which usually points to an issue with permissions or access rights. OK, that's what we thought, which is cool. Uh-huh. Since repositories with ECR are private by default, both you and player A have set up private repositoriesies that no-one else can access until you give someone else specific permissions. Yeah, so what we've set up are private repositories, my friends. Do you see even in the left-hand navigation panel for Maximus, it's images that highlight a thing is under the heading private registry. Um, so what's really happening here is it's so private that no one else, literally no one else can have access until we give it to them specifically. And that's the reason why we can just openly paste the links to our registries in the chat because actually we know that even if you have access to that link, you can't actually get access to the registry itself until we give you specific access. Cool. Let's take a screenshot, so we're gonna go back to our terminal. And just take a screenshot of this error right here. Nice. And I'm gonna paste that into a box here and then move on to the next question.



# Custom Vocabulary

We can resolve transcription inaccuracies using a custom vocabulary, which is a defined list of technical or misspelt words that we want Transcribe to process accurately. A custom vocabulary improves accuracy by defining the correct transcription.

To create an item in a custom vocabulary, you need to define two values. They are 'Phrase', which means the phrase we want Transcribe to detect; and 'DisplayAs', which is the way we want Transcribe to display the word in its transcriptions.

Our custom vocabulary defines the proper spelling/treatment of three inaccuracies in our baseline transcription. First, 'player' -> 'Player'. Second, '4 or 3 forbidden' -> '403 Forbidden'. Third, 'repositoriesies' -> 'repositories'.

**View and edit vocabulary - new (3)** [Info](#)

Reset vocabulary

Delete

Download latest ▼

Q Filter phrase, SoundsLike, IPA or DisplayAs

Show all ▼

< 1 >

| <input type="checkbox"/> | Phrase <a href="#">↗</a> ▼ | SoundsLike - optional <a href="#">↗</a> ▼ | IPA - optional <a href="#">↗</a> ▼ | DisplayAs - optional <a href="#">↗</a> ▼ |
|--------------------------|----------------------------|-------------------------------------------|------------------------------------|------------------------------------------|
| <input type="checkbox"/> | player                     | -                                         | -                                  | Player                                   |
| <input type="checkbox"/> | repositoriesies            | -                                         | -                                  | repositories                             |
| <input type="checkbox"/> | four-or-three-forbidden    | -                                         | -                                  | 403 Forbidden                            |

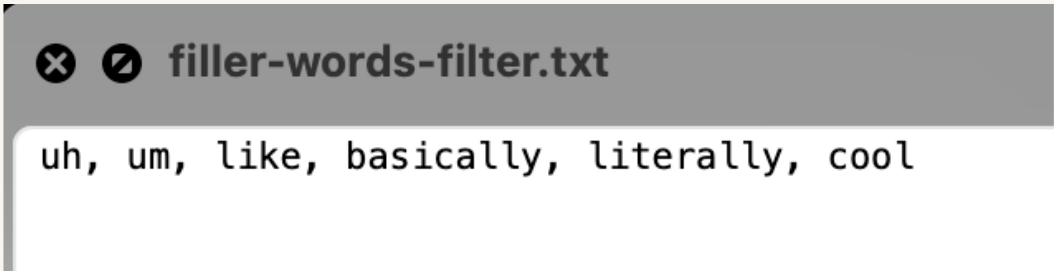
Add row



# Vocabulary Filters

Another feature in Transcribe is vocabulary filtering, which is a tool for removing unwanted words. It's different from custom vocabularies - vocabularies are used for words we DO want to transcribe (accurately), whereas filtered words aren't wanted.

Our vocabulary filter removes unwanted text, like filler words ('um'). To set up this filter, we first created a filter file, which is a comma separated list of words.



✕ 🔒 filler-words-filter.txt

uh, um, like, basically, literally, cool



# Enhanced Transcription

## I ran a new transcription with my custom vocabulary and filtering settings

The enhanced transcription is better than the baseline because it does not have the same typos; it's now capitalised 'player' to turn into a title, and filler words are correctly identified and masked! We'll need more training to fix the '403' error.

| Text                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Audio identification | Subtitles | Toxicity detection - new |
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# Real Time Transcription

For our project extension, we experimented with real-time transcription, which is transcription that happens live while the speaker is still talking. This is helpful for apps that want to deliver live captioning or voice commands.

Even during real-time transcription, we could use features like vocabulary filtering and custom vocabulary. Compared to a transcription job, real-time transcription was still just as accurate, although `403 Forbidden` still has errors.

## Real-time transcription [Info](#)

See how Amazon Transcribe creates a text copy of speech in real time. Choose **Start streaming** and talk.

### Transcription

[Download full transcript](#)

[Start streaming](#)

### Transcription output

Current language: English, US

But that's OK. Let's start.  
Hi, my name is Nick work. I am Player A in a project on Docker and containers. \*\*\*, there was a 403 forbidden response.



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